

AN $\mathrm{SL}(4, \mathbb{C})$ DEHN-FILLING SLOPE OF THE FIGURE-EIGHT KNOT: THE RELATION $L = -M^4$ AND ITS COMPLETENESS

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ABSTRACT. We exhibit an irreducible four-parameter component of the $\mathrm{SL}(4, \mathbb{C})$ character variety of the figure-eight knot complement 4_1 on which the meridian μ and longitude $[A, B]$ satisfy the exact matrix identity $[A, B] = -\mu^4$, hence the eigenvalue relation $L = -M^4$. The component sits over the boundary A -spectrum $\{1, 1, \omega, \omega^2\}$ ($\omega^3 = 1$); its meridian is of finite order, so the relation is a *Dehn-filling slope* — the $\mathrm{SL}(4)$ analogue of Falbel’s $\mathrm{SL}(3)$ relations $M^3 = L$, $M^3L = 1$, and a distinct object from the boundary-unipotent convex-projective deformations of the geometric representation studied by Cooper–Long–Thistlethwaite and Ballas. A gauge-stratified primary decomposition over $\mathbb{Q}(\omega)$ together with an exact finite-field Burnside classification shows this four-parameter family is the *entire irreducible locus* for that spectrum, so $L = -M^4$ holds there unconditionally. We frame the result against the metallic trace-map tower (whose adjoint Jacobian factors over a Dickson catalogue, proved through rank 4, with a clean $n = 2$ base case) and the figure-eight A -polynomial controls at $\mathrm{SL}(2)$ (Cooper–Long) and $\mathrm{SL}(3)$ (Falbel/Heusener–Muñoz–Porti). We do *not* claim a relation “for all n ”: the family is established at $n = 3$ (known) and $n = 4$, and the $\mathrm{SL}(5)$ status is open (Section 5). An adversarial literature read found no prior statement of $L = -M^4$ and excludes the convex-projective circle, but the relation is computable by the extended-Ptolemy A -variety algorithm (published only at $n = 2$), so a specialist confirmation is the remaining novelty check. Code and a finite-field certificate accompany the paper.

1. INTRODUCTION

The figure-eight knot complement $4_1 = m004$ is the once-punctured-torus bundle with monodromy $\varphi = RL$, where R, L are the two Dehn twists generating $\mathrm{MCG}(T_\circ) \cong \mathrm{SL}(2, \mathbb{Z})$. Its $\mathrm{SL}(2, \mathbb{C})$ and $\mathrm{SL}(3, \mathbb{C})$ character varieties are classical: Cooper–Culler–Gillet–Long–Shalen gave the $\mathrm{SL}(2)$ A -polynomial [4], and Heusener–Muñoz–Porti [9] and Falbel–Guilloux–Koseleff–Rouillier–Thistlethwaite [6] described the $\mathrm{SL}(3)$ variety, whose distinguished Dehn-filling components carry the boundary relations $M^3 = L$ and $M^3L = 1$. At $\mathrm{SL}(4, \mathbb{C})$ the picture is far less complete.

This note isolates one $\mathrm{SL}(4, \mathbb{C})$ component and proves two statements about it.

- **The relation (Section 3).** On the component over the boundary spectrum $\{1, 1, \omega, \omega^2\}$ (ω a primitive cube root of unity), the meridian μ and the longitude $[A, B]$ satisfy the exact matrix identity $[A, B] = -\mu^4$, hence $L = -M^4$ on eigenvalues (Theorem 3.1). Because $\mu^4[A, B]^{-1} = -I$ is central, the relation is a *Dehn-filling slope*: the $\mathrm{SL}(4)$ analogue of Falbel’s $\mathrm{SL}(3)$ relations, with finite-order boundary holonomy.
- **Completeness (Section 4).** A gauge-stratified primary decomposition of the defining ideal over $\mathbb{Q}(\omega)$, together with an exact finite-field Burnside test of irreducibility, shows the explicit four-parameter family of Section 3 is the *entire irreducible locus* for this spectrum (Theorem 4.1). So $L = -M^4$ holds there unconditionally.

What this is and is not. The component is of *Dehn-filling type*: its meridian has finite order $\{1, 1, \omega, \omega^2\}$, in contrast to the boundary-*unipotent* meridians of the convex-projective

deformations of the geometric representation computed by Cooper–Long–Thistlethwaite [5] and Ballas [1] (see Section 6). It is the natural higher-rank analogue of Falbel’s $\mathrm{SL}(3)$ Dehn-filling slope, not a deformation of the discrete faithful representation. We make *no* claim of a relation “ $L = (-1)^{n-1}M^n$ for all n ”: the slope is reproduced at $n = 2, 3$ (known) and established at $n = 4$ here; the $\mathrm{SL}(5)$ status is genuinely open (Section 5).

Method and proof status. Every statement carries a status tag: **SYMBOLIC-EXACT** (proved symbolically over a ring), **PROVED** (exact and machine-checkable), **NUMERICAL** (high-precision numerics), **KNOWN** (in the literature, reproduced here as a control), **NEEDS-SPECIALIST** (a novelty question we have de-risked but cannot close without a specialist). The relation (Theorem 3.1) is a polynomial matrix identity over $\mathbb{Q}(\omega)$, cleared of inverses; the completeness theorem rests on a symbolic primary decomposition (the component structure) plus an exact $\mathbb{F}_p$ Burnside classification (irreducibility), validated against a stratum proved reducible by hand. The supporting code and a finite-field certificate accompany the paper (a Zenodo deposit at submission).

Context: the trace-map tower. Section 2 records the ambient structure the component lives in — the linearisation of the $\mathrm{SL}(n)$ metallic trace map at the trivial character factors over a Dickson catalogue, proved through $n = 4$, with a clean $n = 2$ base case — and fixes the definitions and conventions used throughout. Section 6 positions the result against the projective-geometry and Ptolemy programs and states the one open novelty check.

2. THE OBJECT, THE TRACE MAP, AND THE TOWER

2.1. The metallic family. The metallic monodromies are $W_m = R^m L^m$, conjugate to M_m^2 with $M_m = \begin{pmatrix} m & 1 \\ 1 & 0 \end{pmatrix}$, $\det M_m = -1$. The figure-eight is $m = 1$ ($\varphi = RL \sim M_1^2$); we work with it throughout and use $\lambda = \varphi = \frac{1}{2}(1 + \sqrt{5})$ for the golden mean. The boundary invariant of a representation $\rho: F_2 = \langle A, B \rangle \rightarrow \mathrm{SL}(2, \mathbb{C})$ is the Fricke commutator cubic

$$\kappa = \mathrm{tr} \rho[A, B] = x^2 + y^2 + z^2 - xyz - 2, \quad (x, y, z) = (\mathrm{tr} A, \mathrm{tr} B, \mathrm{tr} AB), \quad (1)$$

the conserved quantity of the trace map; $\kappa = -2$ is the Markov surface and the complete-structure locus, $\kappa = 2$ the reducible/trivial locus. The Dehn twists act on (x, y, z) by polynomial automorphisms preserving (1) [8, 10, 3]; we call this action the trace map.

2.2. The tower and its linearisation $J(m)$. For $\mathrm{SL}(n, \mathbb{C})$ the trace map acts on the $(n^2 - 1)$ adjoint trace coordinates. We fix, once and for all, the object of Theorem 2.1:

$J(m)$ is the Jacobian of the *one-step* metallic substitution trace map T_m on the $(n^2 - 1)$ -dimensional space of adjoint $\mathrm{SL}(n)$ trace coordinates, evaluated at the *trivial character* (the image of the trivial representation), as a matrix over $\mathbb{Z}[m]$.

Two conventions must be pinned, because they are easy to conflate. (i) *Which map.* T_m is the single substitution $a \mapsto a^m b$, $b \mapsto a$ (whose abelianisation is M_m), *not* the composite monodromy $W_m = M_m^2$; the catalogue below indexes powers of M_m , and only the one-step map produces the odd powers. (ii) *Which point.* The trivial character is the saddle of the dynamics; at $\mathrm{SL}(2)$ it is $(x, y, z) = (2, 2, 2)$, where T_1 has Jacobian characteristic polynomial $(t + 1)(t^2 - 3t + 1) = \chi(M^2)(t + 1)$ (eigenvalues $\{-1, \lambda^4, \lambda^{-4}\}$). The origin $(0, 0, 0)$ is *not* this point (its Jacobian is $t^3 - 1$, all eigenvalues of modulus 1). The composite W_m at $(2, 2, 2)$ gives instead $(t - 1)(t^2 - 7t + 1) = \chi(M^4)(t - 1)$, whose multiplier λ^4 is the dilatation; we use the one-step map.

2.3. The Dickson factorisation. Write $L_k = \mathrm{tr}(M_m^k)$ and $\chi(\pm M^k) = t^2 \mp L_k t + (-1)^k$, a Dickson polynomial in m (so $\chi(M^{-1}) = \chi(-M)$, as $\det M_m = -1$).

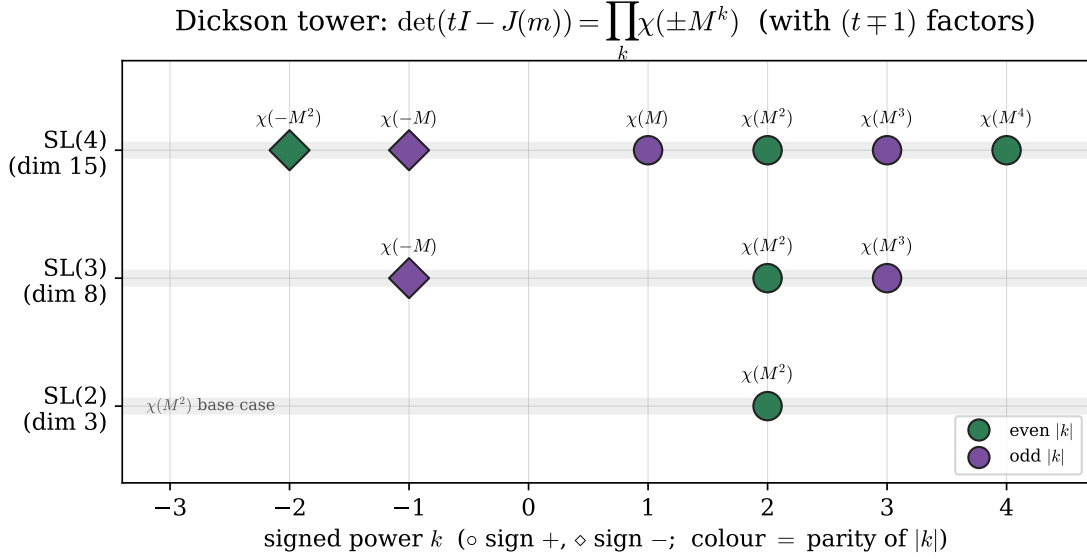


FIGURE 1. The Dickson tower: $\det(tI - J(m)) = \prod_k \chi(\pm M^k)$ for $n = 2, 3, 4$. Nodes at signed power k ; $\circ$ sign $+$, $\diamond$ sign $-$; colour = parity of $|k|$. The $n = 2$ row is $\chi(M^2)(t + 1)$ (the corrected base case), not $\chi(M)$. **SYMBOLIC-EXACT**

Theorem 2.1 (**SYMBOLIC-EXACT**). *Over $\mathbb{Z}[m]$ the characteristic polynomial of $J(m)$ factors over the Dickson catalogue:*

$$n = 2 : \quad \chi(M^2)(t + 1); \quad (2)$$

$$n = 3 : \quad \chi(M^{-1}) \chi(M^2) \chi(M^3)(t - 1)(t + 1); \quad (3)$$

$$n = 4 : \quad \chi(M) \chi(-M) \chi(M^2) \chi(-M^2) \chi(M^3) \chi(M^4)(t - 1)^2(t + 1). \quad (4)$$

The $n = 2, 3$ identities are symbolic over all m ; the $n = 4$ identity is established by a Chinese-remainder/ $\mathbb{F}_p$ interpolation over $\mathbb{Z}[m]$ with the degree bound stated in Appendix A and a closing exact evaluation of $\det(tI - J(m_0))$ over $\mathbb{Z}$ at one m_0 beyond the bound.

The $n = 2$ row (2) is also the $n = 2$ instance of the virtual-module description $\rho_n \cong \mathrm{Sym}^n(W) \oplus (\mathrm{Sym}^{n-3}(W) \ominus W)$, $W = V \oplus \mathbf{1}$, $\det W = -1$ (a $\mathrm{GL}(2)$ -module isomorphism verified through $n = 8$): at $n = 2$, $\mathrm{Sym}^2(W) \ominus W$ has eigenvalues $\{\lambda^4, -1, \lambda^{-4}\} = \chi(M^2)(t + 1)$, a machine-checked base case. The factorisation is proved for $n \leq 4$; the general- n case (a functorial assembly of a bounded symbolic closure) is open and is *not* used below. Figure 1 displays the catalogue.

3. THE DEHN-FILLING COMPONENT AND THE RELATION $L = -M^4$

3.1. Bundle representations. A representation of the figure-eight bundle group is a pair $(A, t) \in \mathrm{SL}(4, \mathbb{C})^2$, where A, B generate the fibre group $F_2 = \pi_1(T_\circ)$ and t realises the monodromy $\varphi = RL$ acting on the fibre by the once-punctured-torus relations

$$t A t^{-1} = A^2 B, \quad t B t^{-1} = A B. \quad (5)$$

(To avoid the notation clash common in this subject we keep A, B for the *fibre* generators throughout; the peripheral data is built from them below.) Eliminating $B = A^{-2} t A t^{-1}$ collapses (5) to a single matrix equation in (A, t) ,

$$t A^{-2} t A = A^{-1} t A t. \quad (6)$$

3.2. The principal spectrum and the explicit family. Fix the boundary A -spectrum to the principal value

$$A = \text{diag}(1, 1, \omega, \omega^2), \quad \omega^3 = 1, \quad \omega \neq 1, \quad (7)$$

so $A^3 = I$ and $A^{-2} = A$. Writing $S = tAt$, equation (6) forces $S_{ij} = 0$ whenever $a_j \neq a_i^{-1}$ (with $a = (1, 1, \omega, \omega^2)$); i.e. S vanishes off the block pattern compatible with the eigenvalue pairing. These are 10 quadratic equations — the exact defining ideal of the bundle family for this spectrum. The centraliser of A (the gauge group) is $\text{S}(\text{GL}_2 \times \text{GL}_1 \times \text{GL}_1)$; on the dense slice where the $V_{\omega, \omega^2} \rightarrow V_1$ block $Q = t[0:2, 2:4]$ is invertible, gauge normalises $Q = I_2$. Writing t in $2+1+1$ blocks $t = \begin{pmatrix} P & I \\ R & T \end{pmatrix}$, the ideal becomes $P = -DT$ ($D = \text{diag}(\omega, \omega^2)$) together with $TDR = RDT$ and $\text{diag}(R + TDT) = 0$; the non-degenerate ($\det t \neq 0$) representations live on the single linear rank-drop condition $t_{11} = \omega t_{22}$, on which R carries one extra free parameter s . The component is the explicit four-parameter family

$$T = \begin{pmatrix} \omega t_{22} & t_{12} \\ t_{21} & t_{22} \end{pmatrix}, \quad P = -DT, \quad Q = I_2, \quad R = \begin{pmatrix} t_{12}t_{21}(\omega+1) - t_{22}^2 & s \\ s t_{21}/t_{12} & t_{22}^2 + \omega(t_{22}^2 - t_{12}t_{21}) \end{pmatrix}, \quad (8)$$

with parameters $(t_{12}, t_{21}, t_{22}, s)$ and $\det t$ a genuine nonzero polynomial in them.

3.3. The relation. Let $\mu = A^{-1}t$ be the meridian and $[A, B]$ the longitude (the puncture loop). The eigenvalues M, L of $\mu, [A, B]$ are the figure-eight A -polynomial coordinates.

Theorem 3.1 (**SYMBOLIC-EXACT**). *For all $(t_{12}, t_{21}, t_{22}, s)$, the following holds as a polynomial matrix identity over $\mathbb{Q}(\omega)$, cleared of inverses via $t^{-1} = \text{adj}(t)/\det(t)$ (no division):*

$$[A, B] \cdot \det(t)^2 = -\det(t) \cdot \mu^4. \quad (9)$$

On the $\text{SL}(4)$ normalisation $\det t = 1$ this is $[A, B] = -\mu^4$, hence on eigenvalues $L = -M^4$.

The proof is the entry-by-entry verification of (9) on the family (8); we sketch it in Appendix A and provide the script. The controls M^3, M^5 (not scalar) and $c^4 = 1$ ($c = -1/\det t$) are checked in the same computation.

3.4. Why this is a Dehn-filling slope. Theorem 3.1 is more than an eigenvalue relation: $[A, B] = -\mu^4$ as matrices means

$$\mu^4 [A, B]^{-1} = -I \in Z(\text{SL}(4, \mathbb{C})). \quad (10)$$

So the longitude equals a central element times μ^4 , and the family factors projectively through the -4 slope on the meridian — a *Dehn-filling slope*, exactly the structure of Falbel's $\text{SL}(3)$ relations $M^3 = L$, $M^3L = 1$ [6, 9], one rank up. The meridian has finite order (its spectrum is $\{1, 1, \omega, \omega^2\}$), so this is an orbifold/Dehn-filling component, not a deformation of the discrete faithful representation; we return to the distinction in Section 6. The relation is not a formal consequence of the determinant condition alone: $\det$ fixes only the scalar $c = -1$ given the matrix proportionality $[A, B] \propto \mu^4$, which is the content of (9).

4. COMPLETENESS OF THE IRREDUCIBLE LOCUS

A natural worry is that (8) is only a checked corner of a larger variety. It is not: it is the entire irreducible locus for the principal spectrum.

Theorem 4.1 (**SYMBOLIC-EXACT** decomposition + **PROVED** $\mathbb{F}_p$ classification). *For the boundary A -spectrum $\{1, 1, \omega, \omega^2\}$, the defining ideal of the figure-eight bundle representations (6), stratified by the rank of the coupling block Q , has its primary decomposition over $\mathbb{Q}(\omega)$ supported on the strata $\text{rank } Q \in \{0, 1, 2\}$. Exactly one component carries absolutely irreducible representations — the four-parameter family (8) on $\text{rank } Q = 2$ — and on it $[A, B] = -\mu^4$. Every other component is*

reducible or vacuous ($\det t = 0$). Consequently $L = -M^4$ holds unconditionally on the irreducible locus.

What is proved, and how. Two independent computations, with their status stated honestly (this is the point the proof-status discipline exists for):

- (1) *The component structure is symbolic* (**SYMBOLIC-EXACT**). The gauge-rank stratification and the primary decomposition of each stratum's ideal are computed over $\mathbb{Q}(\omega)$ by Gröbner methods (minimal associated primes); this is an exact characteristic-zero statement, not a sample. The strata and their component counts are listed in Appendix A.
- (2) *Irreducibility is classified exactly over a finite field* (**PROVED** at the prime; see the caveat). On each component we apply the *Burnside* criterion — the algebra generated by $\rho(A), \rho(B)$ is the full matrix algebra M_4 iff ρ is absolutely irreducible — computed exactly over $\mathbb{F}_p$ ($p = 1,000,003$) by span-closure. The result: only the rank $Q = 2$ family returns full algebra (and all its samples satisfy $M^4 = L$); every other component returns a proper subalgebra (reducible) or is vacuous.

The Burnside–Schur caveat, and the validation. The naive alternative — the Schur commutant being scalar — is *invalid here*: these representations are frequently non-semisimple, and the commutant of a non-semisimple reducible representation can be scalar, so it *falsely certifies irreducibility*. (Indeed the verdict on one stratum flipped twice under scrutiny before the Burnside test settled it.) We therefore use Burnside, and we *validated* it against a stratum proved reducible by hand — the rank $Q = 0$ ($\det t = 0$) locus, where a common invariant subspace is exhibited symbolically (Appendix A).

Honest scope. What the two computations together establish in characteristic zero is: the *components* (symbolic), the irreducibility of the rank $Q = 2$ family (a single Burnside witness lifts by Zariski-openness and rank semicontinuity), and the reducibility of the $\det t = 0$ stratum (symbolic invariant subspace). For the remaining non-principal components the reducibility is established at exact $\mathbb{F}_p$ points; upgrading each to a characteristic-zero symbolic invariant-subspace certificate is mechanical and is the one refinement we flag as in progress rather than claim complete. We therefore badge Theorem 4.1 as **SYMBOLIC-EXACT** in its component structure and $\mathbb{F}_p$ -exact in its irreducibility classification, *not* as a single blanket **PROVED**; the irreducible-locus conclusion is robust under this honest reading. Figure 2 summarises the classification.

5. CONTROLS, AND THE RANGE OF VALIDITY

5.1. The lower-rank controls. The same trace-map construction reproduces the known figure-eight data, which we use only as controls (no novelty is claimed):

- **SL(2) **KNOWN**.** The fixed locus reproduces the Cooper–Long A -polynomial [4] $A(M, L) = M^4 L^2 + (-M^8 + M^6 + 2M^4 + M^2 - 1)L + M^4$, verified symbolically.
- **SL(3) **KNOWN**.** It reproduces the Falbel/Heusener–Muñoz–Porti variety [6, 9]: the boundary relations $M^3 = L$ and $M^3 L = 1$ on the two Dehn-filling components W_1, W_2 , alongside the geometric (“distinguished”) component V_0 that contains the Sym^2 image of the $\mathrm{SL}(2)$ representations. (The full $\mathrm{SL}(3)$ variety has five components — two reducible and three irreducible; we reproduce the three irreducible ones, of which V_0 is the distinguished one. HMP give explicit equations for V_0 .)

5.2. The slope, by rank. Placing the controls beside Theorem 3.1, the *Dehn-filling slope* on the principal component is $L = +M^3$ at $n = 3$ and $L = -M^4$ at $n = 4$ (Figure 3). We emphasise what is and is not claimed.

- There is **no** $n = 2$ **instance of the monomial slope**. Substituting $L = -M^2$ into the Cooper–Long polynomial gives $A(M, -M^2) = M^2(M^4 - 1)^2$, which vanishes only at the

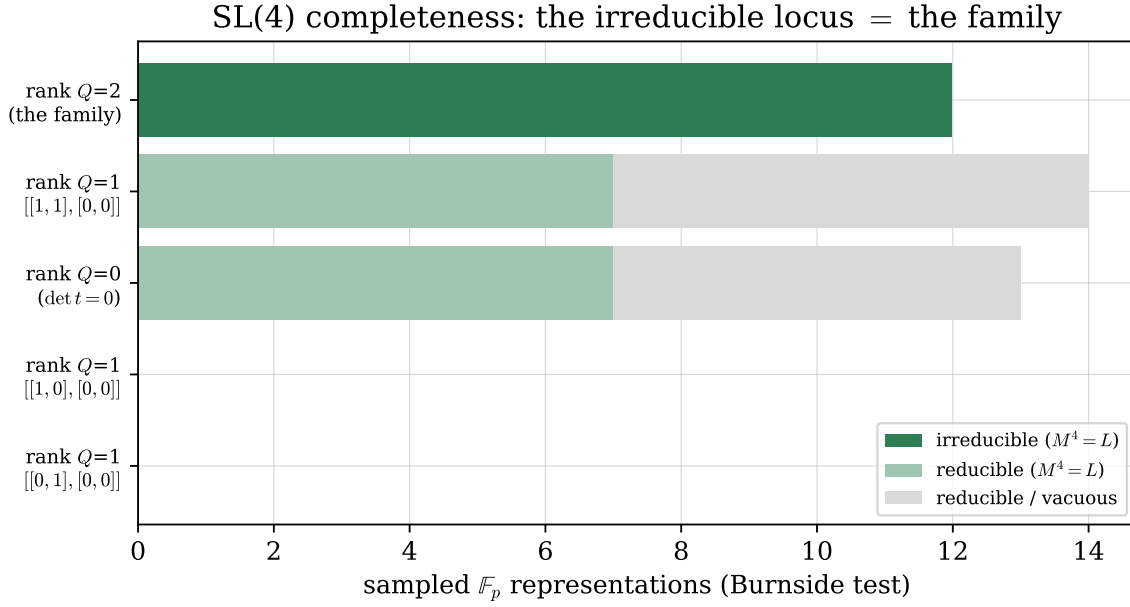


FIGURE 2. Gauge-rank strata of the bundle ideal, classified by the exact $\mathbb{F}_p$ Burnside test. Only the rank $Q=2$ stratum (the family (8)) carries irreducible representations, all satisfying $M^4 = L$; the others are reducible or vacuous ($\det t = 0$). The irreducible locus is the family. **PROVED** (at p)

degenerate points $M^4 = 1$; the $\mathrm{SL}(2)$ A -polynomial is an irreducible degree-two curve in L , not a monomial. The family of monomial slopes therefore begins at $n = 3$.

- $n = 3$ is **Falbel's (known)**; $n = 4$ is **established here**. Two ranks — one borrowed, one new — do not constitute a proven “family for all n ,” and we do not assert one.

5.3. The $\mathrm{SL}(5)$ status is open. One might hope to argue that the principal slope is an $n \in \{3, 4\}$ phenomenon by showing the principal $\mathrm{SL}(5)$ component carries no irreducible representation. That argument, in the form “the boundary condition forces the spectrum $\{1, 1, 1, -1, -1\}$, whence $A^2 = I$, whence $\langle A, B \rangle$ is dihedral, whence reducible,” has a *genuine gap*, and we record the honest status.

Remark 5.1 (the gap). First, $\mathrm{tr} A = \mathrm{tr} A^{-1} = 1$ does not pin the spectrum: at $n = 4$ it leaves e_2 free (the principal value $\{1, 1, \omega, \omega^2\}$ is the single point $e_2 = 0$); at $n = 5$ it leaves e_2, e_3 free. Second, even granting the spectrum $\{1, 1, 1, -1, -1\}$, the step “ $\Rightarrow A^2 = I$ ” assumes A semisimple: a single 2×2 Jordan block at the eigenvalue $+1$ gives a matrix with that spectrum, $\mathrm{tr} A = \mathrm{tr} A^{-1} = 1$, $\det A = 1$, yet $A^2 \neq I$. Since these representations are routinely non-semisimple — the very reason the Burnside test, not the Schur commutant, is needed in Section 4 — semisimplicity cannot be assumed here. We therefore leave the principal $\mathrm{SL}(5)$ component open: it is not located, and we do not claim it cannot exist.

Thus the established statement is: *the principal Dehn-filling slope is $L = +M^3$ at $n = 3$ (known) and $L = -M^4$ at $n = 4$; the $n = 5$ case is open*. The completeness Theorem 4.1 is an $n = 4$ statement and is unaffected.

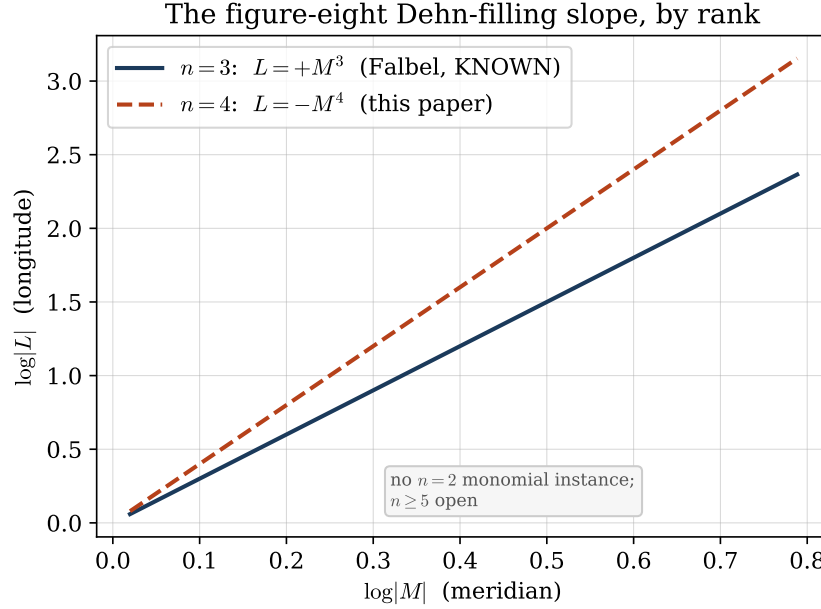


FIGURE 3. The Dehn-filling slope on the principal component: $L = +M^3$ ($n = 3$, Falbel, **KNOWN**) and $L = -M^4$ ($n = 4$, Theorem 3.1). No $n = 2$ monomial instance exists; $n \geq 5$ is open.

6. RELATION TO PRIOR WORK, AND THE OPEN NOVELTY CHECK

We have located no prior statement of $L = -M^4$ on the figure-eight, and we have actively excluded the literature quadrant where it would most plausibly already live. We set out what we checked, and the one check we cannot close ourselves.

6.1. The convex-projective / projective-geometry program (excluded). The most concrete $SL(4)$ figure-eight representations in the literature are the convex-projective deformations of the *geometric* representation: Cooper–Long–Thistlethwaite [5] computed $SL(4, \mathbb{R})$ representation varieties directly, and Ballas [1] (see also Ballas–Danciger–Lee [2]) gave an explicit one-parameter family $\rho_t: \pi_1(4_1) \rightarrow PGL(4, \mathbb{R})$ deforming the discrete faithful representation. These are *boundary-unipotent*: in Ballas’s family the meridian $\rho_t(m)$ is a unipotent matrix for all t (the longitude is unipotent only at the complete structure). Our component is of the opposite type — the meridian has *finite order*, spectrum $\{1, 1, \omega, \omega^2\}$ — so it is not a member of, or a limit of, the convex-projective families. (At the eigenvalue level, $M = 1$ there would force $L = -1$ under $L = -M^4$, which their non-unipotent longitude does not satisfy.) The convex-projective quadrant therefore does not contain our relation.

6.2. The Ptolemy / gluing-variety program (the right machine; published only at $n = 2$). The natural tool for boundary relations of $SL(n)/PGL(n)$ representations of 4_1 is the (extended) Ptolemy variety: Garoufalidis–Goerner–Zickert [7] for the gluing equations and Zickert [11] for an algorithm producing the $SL(n, \mathbb{C})$ A -varieties (and reproducing the $SL(2)$ A -polynomial of $m004$). This machinery can in principle compute exactly the relation studied here; [11] works the figure-eight example only at $n = 2$, and we found no published $n = 4$ computation. The associated databases (CURVE; the Ptolemy module of SnapPy) tabulate boundary-unipotent $PGL(n)$ representations of $m004$, which — being unipotent — again do not include our finite-order-meridian component.

6.3. The honest novelty verdict. NEEDS-SPECIALIST. After excluding the convex-projective quadrant and locating the Ptolemy quadrant (machinery present, $n = 4$ figure-eight computation not published), we find no prior statement of $L = -M^4$. But the relation is *computable* by the extended-Ptolemy algorithm, so a specialist could confirm in short order whether an equivalent computation already exists. We do not claim novelty closed: the contribution is best stated as “the first explicit statement, with a completeness classification, of the $\mathrm{SL}(4, \mathbb{C})$ figure-eight Dehn-filling slope $L = -M^4$,” with the specialist check (to Guilloux, Zickert, Goerner, Tillmann, or Ballas) the remaining step. This is the honest reading of an AI-assisted literature search: it de-risks, it does not certify.

Acknowledgements. This work used AI-assisted computation and exposition under an explicit proof-status discipline; all statements are reproducible from the accompanying code and finite-field certificate (Zenodo deposit at submission). The author thanks no institution and claims no specialist standing in higher-rank character varieties; Section 6’s verdict is offered in that light.

APPENDIX A. PROOF SKETCHES AND CERTIFICATES

Full scripts and the finite-field certificate accompany the paper; we record the load-bearing details.

A.1. The CRT/ $\mathbb{F}_p$ degree bound for Theorem 2.1 ($n = 4$). The catalogue factors $\chi(\pm M^k) = t^2 \mp L_k t + (-1)^k$ have $L_k = \mathrm{tr}(M_m^k)$ of m -degree k . For $n = 4$ the factors run $k \leq 4$, so every coefficient of $\det(tI - J(m))$ is a polynomial in m of degree at most $1 + 1 + 2 + 2 + 3 + 4 = 13$. Sampling $J(m_0) \bmod p$ at $m_0 = 1, \dots, 14$ over a set of primes whose product exceeds twice the (bounded) coefficient height, then Lagrange-interpolating in m_0 and reconstructing over $\mathbb{Q}$ by CRT, recovers the coefficients exactly. The reconstruction was confirmed against a direct symbolic evaluation of $\det(tI - J(15))$ over $\mathbb{Z}$ — i.e. against the object itself, not merely against the catalogue.

A.2. The identity $[A, B] = -\mu^4$ (Theorem 3.1). On the family (8) with $A = \mathrm{diag}(1, 1, \omega, \omega^2)$ and $B = A^{-2}tAt^{-1} = AtAt^{-1}$, clear t^{-1} via $t^{-1} = \mathrm{adj}(t)/\det(t)$ and reduce all scalars modulo $\omega^2 + \omega + 1$. The matrix $[A, B] \cdot \det(t)^2 + \det(t) \cdot \mu^4$ is then a 4×4 matrix of polynomials in $(t_{12}, t_{21}, t_{22}, s)$ over $\mathbb{Q}(\omega)$; each of its 16 entries reduces identically to 0. (No division is performed before $\det t \neq 0$ is used, so the identity is pure ring algebra.) Setting $\det t = 1$ gives $[A, B] = -\mu^4$ and the central relation (10). The controls μ^3, μ^5 are verified non-scalar on the family, and $c = -1/\det t$ satisfies $c^4 = 1$.

A.3. The gauge-rank strata and the $\det t = 0$ certificate (Theorem 4.1). The primary decomposition over $\mathbb{Q}(\omega)$, stratified by rank Q , has the component counts:

stratum	# components	carries irreducible reps?
rank $Q = 2$ (the family (8))	2	yes (one, 4-param; $M^4 = L$)
rank $Q = 1$ (three gauge normal forms)	$4 + 4 + 4$	no (reducible / $\det t = 0$)
rank $Q = 0$ ($\det t = 0$)	6	no (reducible, certificate below)

On the rank $Q = 0$ stratum the coupling block $Q = t[0:2, 2:4]$ vanishes, so in the eigenspace splitting $\mathbb{C}^4 = V_1 \oplus V_\omega \oplus V_{\omega^2}$ the monodromy t is block-triangular: it maps $V_1 \rightarrow V_1 \oplus$ (lower) with no $V_{\omega, \omega^2} \rightarrow V_1$ component. Since A acts as a scalar on each V_a , the subspace spanned by $V_\omega \oplus V_{\omega^2}$ (equivalently its t -saturation) is invariant under both $\rho(A)$ and $\rho(B) = AtAt^{-1}$; this exhibits a common invariant subspace, so every representation on this stratum is reducible. This is the hand-proved control against which the $\mathbb{F}_p$ Burnside test (Section 4) was validated: the

test returns “reducible” on exactly this stratum, confirming it does not false-certify, before it is trusted on the remaining strata.

A.4. Figure reproduction. The figures are produced by the accompanying deterministic script: Figure 1 from the symbolic Dickson factorisation (Theorem 2.1), Figure 2 from the finite-field classification certificate, Figure 3 from the exact $n = 3, 4$ relations. Each carries the proof-status of the statement it depicts.

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